Whani Kim

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Education

Seoul National University
Seoul, South Korea

M.A. in Cognitive Psychology
March 2019 — February 2021

- Advisor: Sowon Hahn, Ph.D.
- Master's Thesis: "The Role of Robot Body Language on the Perception of Social Qualities and Human-likeness in Robots"

University of Texas at Arlington
Arlington, Texas

B.S. in Psychology
August 2011 — May 2016

- Distinguished Scholars Program (Advisor: Shannon Layman, Ph.D.)
- Honors Thesis: "Comfort in Music: Effect of Music on the Need for Cognitive Closure"

Professional Experience

R&D/Product Lead
Seoul, South Korea

HAII Corp.
Oct 2023 — Present

- Spearheaded the drafting, technical planning, and submission of government R&D grant proposals, successfully securing funding for medical device development and clinical demonstration.
- Mentored graduate researchers of Yonsei University's HCI Lab, guiding master's students through topic conceptualization, experimental design, and academic writing in cognitive science and digital health.
- Collaborated with cross-functional teams to advance Alzguard into confirmatory clinical trials for KFDA approval
- Directed the development and cross-platform deployment of "Brain Coach Sammy" for Samsung Smart TVs

R&D/UX Researcher
Seoul, South Korea

HAII Corp.
Dec 2022 — Present

- Led the development of "Digital Assessment of Cognitive Impairment (DACI)"
- Conducted user research with senior demographics, optimizing application UX and interaction accessibility
- Implemented novel diagnostic tasks for Alzheimer's Disease based on empirical evidence, improving diagnostic accuracy

UX Researcher
Seoul, South Korea

Buyrus Design
Feb 2022 — Dec 2022

- Collaborated with North American counterparts and client to coordinate FGD & IDI interview schedules and manage participants that resulted in drop-out rates of 16% in a 12 week diary study
- Analyzed qualitative and quantitative data to produce actionable insights regarding usability of AI Math Education Application targeted towards North American users
- Conducted desk research, expert interviews, focus group workshops, and user shadowing for next generation in-vehicle infotainment system project and autonomous robotaxi project

Graduate Researcher
Seoul, South Korea

Human Factors Psychology Lab (PI: Sowon Hahn, Ph.D.)
Mar 2019 — Feb 2021

- Managed all stages of research grant processes, from application to reporting, ensuring timely execution
- Developed psychological tasks, questionnaires, and models such as measures of social cognition, psychological resilience, and psychological model of emotion based on breathing patterns respectively

Linguist - ROK-US Combined Forces Command
Pyeongtaek, South Korea

Republic of Korea Army
Nov 2016 — August 2018

- Acted as squadron leader and served as a translator for major ROK-US combined exercises

Publications

- Yun, B.H., Ko, H.J.,**Kim, W.**, Park, S.Y., Kim, J.S., Kim, B.R., Lee, J.H., Kim, G.H., & Kim, J.W. (2026). *A Chatbot that Delivers Cognitive Training: Longitudinal Field Study Assessing Cognitive Function and Social Support.* 2nd round revision.
- **Kim, W.**, Kim, J.S., Park, S.Y., Ko, H.J., Yun, B.H., Kim, Y.Y., Kim, D.H., Lim, S.K., Kim, B.R., Lee, J.H., Kim, G.H., & Kim, J.W. (2026). *Substitute or Supplement? The Role of Multimodal Digital Biomarkers in Mobile Cognitive Impairment Assessment Tools.* 2nd round revision.
- Yun, B.H., Kim, B., Ko, H.J., Kim, J.S., Kim, B.R., **Kim, W.**, Park, S.Y., Kim, J.W., Lee, J.H., & Kim, G.H. (2026). Enhancing Cognitive Functions of Older Adults With Software Robot: Longitudinal Exploratory Field Study. *JMIR mHealth and uHealth, 14*(1), e75308.

- Ko, H.J., Kim, J.S., **Kim, W.**, Yun, B.H., Park, S.Y., Kim, D.H., Kwon, U.J., Lim, S.K., Kim, B.R., Lee, J.H., Kim, G.H., & Kim, J.W. (2025). Optimizing mobile cognitive assessment reduces administration time while maintaining screening accuracy in older adults. *Scientific Reports, 15*(1), 30356.
- Park, S.Y., **Kim, W.**, Ko, H.J., Yun, B.H., Lee, J.H., Kim, J.W. (2025). Alzguard: Usability Evaluation of an Early Alzheimer's Screening App for Older Adults. *Proceedings of HCI Korea 2025 (pp. 1222-1225).*
- Yun, B.H., **Kim, W.**, Ko, H.J., Park, S.Y., Kim, J.S., Kim, G.H., Kim, B.R., Lee, J.H., & Kim, J.W. (2024). Development and effectiveness of an AI chatbot-based mobile cognitive screening and customized training application for preventing dementia: Older adults living in rural areas of South Korea. *Archives of Design Research, 37*(5), 77-90.
- Song, H., Yun, B.H., **Kim, W.**, Ko, H.J., Han, S., Park, S.Y., Kwon, U.J., Yoo, J.S., Kim, B.H., Kim, J.W. (2024). A Study on UX Design of Digital Healthcare Services for the Elderly: Focusing on Cognitive Function Assessment Application 'DACI'. *Proceedings of HCI Korea 2024 (pp. 521-528).*
- Yun, B.H., **Kim, W.**, Ko, H.J., Park, S.Y., Song, H., Noh, M., You, H.J., Kwon, U.J., Yoo, J.S., & Kim, J.W. (2024). Evaluation of the Alztalk A&T Application for Screening Cognitive Function and Customized Training of the Elderly. *Proceedings of HCI Korea 2024 (pp. 601-607).*
- Zhang, X., Lee, S.K., **Kim, W.,** & Hahn, S. (2021). "Sorry, It Was My Fault": Repairing Trust in Human-Robot Interactions. *International Journal of Human-Computer Studies, 175*, 103031

Poster Presentations

- Shim, G., **Kim, W.**, Yun, B.H., Ko, H.J., Park, S.Y., Kim, J.S., Kim, Y.Y., ... Lee, J.H. (2026). Digital Rhythmic Tapping Biomarkers for Early MCI Screening: A Micro-behavioral Analysis. Presented at the *Alzheimer's Association International Conference 2026.* London, United Kingdom.
- Yun, B.H., Kim, B.H., **Kim, W.**, Ko, H.J., Shim, G., Park, S.Y., Kim, J.S., Kim, Y.Y., ... Kim, J.W. (2026). Determinants of Completion and Dropout in a Community-Based Smartphone Cognitive Intervention for Older Adults. Presented at the *Alzheimer's Association International Conference 2026.* London, United Kingdom.
- Kim, J.S., **Kim,W.**, Ko, H.J., Yun, B.H., Park, S.Y., Kim, B.R., ... Kim, J.W. (2025). From Prodromal to Preclinical: Detecting Amyloid-Beta Through Mobile Gaze Analysis. *Alzheimer's Dementia, 21*, e102761.
- Park, S.Y., Lee, J.H., **Kim, W.**, Ko, H.J., Yun, B.H., Kim, J.S., ... Kim, J.W. (2025). A Preliminary Investigation into How Free Speech Tasks Help Detect People Who Are at Risk of Mild Cognitive Impairment. *Alzheimer's Dementia, 21*, e102772.
- **Kim, W.**, Kim, J.S., Ko, H.J., Yun, B.H., Kim, Y., Kim, D.H., ... Kim, J.W. (2024). Investigating the Validity of Digital Biomarkers in the Preclinical Screening of Cognitive Normal, Amyloid-β- MCI, and Amyloid-β+ MCI. *Alzheimer's Dementia, 20*, e090074.
- **Kim, W.**, Kim, J.S., Ko, H.J., Yun, B.H., Kim, Y.Y., Kim, D.H., ... Kim, J.W. (2023). A preliminary feasibility study on screening cognitive impairment based on multi-modal biomarkers and stacking ensemble AI approach. *Alzheimer's Dementia, 19*, e082466.
- Rhee, Y.W., **Kim, W.** &, Hahn, S. (2019). The Role of Trust in Interaction with AI Agents. Presented at *the 60th Annual Meeting of the Psychonomic Society*. Québec, Canada.
- Lee, J., **Kim, W.** &, Hahn, S. (2019). The Effect of Computer Self-Efficacy on Reliance in Automated Car Task. Presented at *the Annual Meeting of the Cognitive Science Society.* Seoul, South Korea
- **Kim, W.** &, Layman, S. (2016). Comfort in Music: Effect of Music on the Need for Cognitive Closure. Presented at *the Annual Celebration of Excellence by Students*. Arlington, Texas

Research Grants and Awards

- **Project Manager**, Development and Validation of a Digital Biomarker-Based Digital Therapeutic Device for Monitoring and Treatment of Age-Related Sarcopenia. *National IT Industry Promotion Agency*, Jan 2026 — Dec 2027
  - Leading the clinical validation and execution of a multi-center confirmatory randomized controlled trial(RCT) for the diagnosis and treatment of sarcopenia
- **Project Manager**, Development of a Next-Generation AI-Driven Full-Cycle Dementia Diagnosis and Biomarker Analysis Solution. *Ministry of SMEs and Startups*, Aug 2025 — Dec 2025
  - Spearheaded research roadmap, defining data collection protocols and clinical objectives.

  - Coordinated clinical data collection with Chonnam National University Hospital, providing oversight to secure high-quality data from 100 patients.
- **2nd Place**, KHIDI Medtech Open Innovation New Impact Project 2024 (Roche Diagnostics Korea). *Korea Health Industry Development Institute*, Jul 2024
  - Recognized for technological capabilities, commercial viability, and global expansion potential
  - Established strategic partnership with Roche Diagnostics Korea to advance global market entry and solution scalability
- **Researcher**, Demonstration Project for the Implementation and Dissemination of AI-Based Digital Health Systems for Speech and Cognitive Rehabilitation in Medical Institutions. *National IT Industry Promotion Agency*, Jan 2024 — Dec 2025
  - Managed a multi-year, multi-center clinical validation project for AI-driven speech and cognitive rehabilitation devices across 7 public hospitals, securing 160+ patient enrollments
  - Developed operational frameworks and data collection workflows to facilitate continuous real-world utilization of AI medical devices in clinical settings
- **Project Manager**, Approval and commercialization of digital medical devices for diagnosis and treatment of cognitive impairment based on artificial intelligence. *Gangwon Information & Culture Industry Promotion Agency*, Jul 2023 — Dec 2025
  - Drove the clinical validation process for a multi-center confirmatory randomized controlled trial (RCT) in dementia diagnosis
  - Developed new cognitive training programs in coordination with Ewha Women's University Hospital
- **Fiscal Manager & Researcher**, Development of AI-based Service Platform for Psychological Self-Test and Management. *Ministry of SMEs and Startups*, Dec 2019 — Dec 2020
  - Managed research budget to ensure compliance with regulations
  - Developed a psychological model of emotion based on breathing patterns
  - Created application wireframes to guide the development process
- **Project Manager**, Analysis of the Effect of Newspaper Reading on Cognitive Processes of the Brain Using EEG. *Korean Newspaper Association*, May 2019 — Nov 2019
  - Managed research operations, protocol development, EEG data acquisition, and team coordination.
  - Developed Korean versions of social cognition tasks: Reading the Mind in the Eyes & Faux Pas Recognition
  - Article on research reported on various major news media outlets
- **Researcher**, Development of Immersive Tourism Technology Utilizing 4D Technology and Affective Modeling. *Korea Culture and Tourism Institute*, Mar 2019 — Dec 2019
  - Developed user interface and user experience in tourist programs to enhance immersiveness
  - Assisted in setting up the programs on site
  - Assisted in gathering data & drafting the final report

Teaching Experience

Interaction Design
Yonsei University

Guest Lecturer
September 2026

- Gave special lecture on the topic of "Psychology in UX"
- For full lecture video - Click here
- For lecture slides - Click here

Introduction to Psychology - International Summer Program
Seoul National University

Teaching Assistant
Summer 2019

Fellowships and Scholarships

College of Social Sciences
Seoul National University

Future Social Sciences Researcher Development Grant
April 2020 — February 2021

- Received ₩3,000,000 ($2,661 equivalent) for Research on "Role of Robot Body Language on the Perception of Physically Embodied Robots"

Woosan Scholarship
Seoul, South Korea

Recipient
March 2019 — February 2021